StorCLI Diagnostics & Administration Guide

Enterprise Technical Manual: MegaRAID Controllers, Physical Disk Diagnostics, Cache Tuning & Automation

Utility: storcli / storcli64	**Target HW:** LSI / Broadcom SAS, SATA, NVMe MegaRAID & HBA	**Platforms:** Linux (Debian/RHEL/Proxmox), ESXi

1. StorCLI Syntax, Hierarchy & Global Output Modes

StorCLI targets hardware hierarchically: storcli /cx[/ex]/sx|vx|dx action [options].

Target Notation	Hardware Level & Target Scope	Standard Diagnostic Command Example
/c0 or /call	Target specific controller (0) or all controllers	`storcli /c0 show status`
/e252 or /eall	Target backplane enclosure (e.g. Enclosure 252)	`storcli /c0/eall show`
/s0 or /sall	Target physical drive slot (e.g. Slot 0) or all slots	`storcli /c0/e252/s0 show all`
/v0 or /vall	Target Virtual Drive (RAID array) or all arrays	`storcli /c0/v0 show all`
/d0 or /dall	Target Drive Group (disk pool)	`storcli /c0/d0 show`
/fall	Foreign RAID configurations across all drives	`storcli /c0/fall show`
/cv or /bbu	CacheVault flash module or Battery Backup Unit	`storcli /c0/cv show all`

PRO TIP: Append **J** to any command (e.g., storcli /c0 show J) for clean JSON output suitable for automated Python/Bash monitoring scripts.

2. Controller Health, Diagnostics & Alarm Management

Command	Diagnostic Purpose & Expected Output
`storcli show`	Enumerate all installed controllers, model names, PCI slots, firmware versions, and overall status.
`storcli /c0 show status`	Summary controller status, memory size, BIOS/firmware versions, and BBU status.
`storcli /c0 show health`	Displays controller operational health (Healthy, Degraded, Failed) and fault conditions.
`storcli /c0 show all`	Comprehensive controller dump: ROC temperature, memory error counts, PHY errors, PCIe link speed.
`storcli /c0 show events filter=warning,error,critical`	Inspect filtered event log: captures disk dropouts, rebuild events, parity errors, and bus resets.
`storcli /c0 show termlog`	Displays live firmware terminal boot/runtime diagnostics log (essential for controller crash debugging).
`storcli /c0 show alarms`	Check status of the physical hardware audible chassis alarm buzzer.
`storcli /c0 set alarm=silence`	Mutes / silences the active sounding hardware buzzer during an ongoing drive failure.
`storcli /c0 set alarm=off`	Permanently disables the controller audible alarm buzzer.
`storcli /c0 flushcache`	Flushes dirty cache blocks to disk before shutdown or maintenance.

3. Physical Disk Diagnostics, Error Counters & SMART Health

Physical disk diagnostics must be evaluated regularly to detect bad sectors and prevent multi-drive failures.

Command	Operational Function	Key Diagnostic Output
`storcli /c0/eall/sall show`	List all physical drives across all slots	State (Onln, UGood, UBad, Failed), Type, Capacity.
`storcli /c0/eX/sX show all`	Detailed diagnostic page for drive in Encl X, Slot X	Media/Other error counters, SMART alerts, Temp, SN.
`storcli /c0/eX/sX show smart`	Query drive internal SMART attributes	SMART Alert Trip: **No** (Healthy) / **Yes** (Failing).
`storcli /c0/eX/sX show error`	Display hardware error statistics recorded by firmware	Media Error Count, Other Error Count, Predictive Fail.

Diagnostic Counter	Threshold	Meaning & Recommended Engineering Action
Media Error Count	**0**	Uncorrectable bad sectors on disk media. Any value > 0 indicates surface degradation. Schedule replacement.
Other Error Count	**0**	Communication errors between drive & backplane (SAS PHY CRC errors, bad slot, or cabling noise).
Predictive Failure Count	**0**	SMART internal algorithms indicate imminent disk crash. Replace drive immediately before array breaks.
Drive Temperature	< 50°C	Operating temperature. Temperatures sustained > 55°C degrade flash/bearings and require cooling review.

4. Physical Drive Identification (LED), Slot States & Modes

Task / Action	Command Syntax	Operational Function & Safety Note
Blink Drive LED (Locate)	`storcli /c0/eX/sX start locate`	Flashes drive tray LED to physically identify the target disk in server chassis.
Stop Blinking LED	`storcli /c0/eX/sX stop locate`	Stops flashing locator LED once disk is identified.
Set Drive Offline	`storcli /c0/eX/sX set offline`	Safely unmounts and isolates drive from active RAID array before physical extraction.
Mark Drive Missing	`storcli /c0/eX/sX set missing`	Marks offline slot as missing so array can prepare for replacement insertion.
Spin Down (Safe Removal)	`storcli /c0/eX/sX spin down`	Parks disk actuator arms and spins down spindle before hot-pulling.
Fix Unconfigured Bad	`storcli /c0/eX/sX set good [force]`	Transitions drive from **UBad** (Unconfigured Bad) to **UGood** (Unconfigured Good).
Set Pass-Through (JBOD)	`storcli /c0/eX/sX set jbod`	Sets drive to raw pass-through mode (required for ZFS, Ceph, and software-defined storage).
Set RAID Mode (Good)	`storcli /c0/eX/sX set good`	Reverts JBOD drive to Unconfigured Good state for RAID array assignment.

5. Virtual Drive (VD) Diagnostics & Array Operations

Action	Command Syntax	Description & Details
List All Virtual Drives	`storcli /c0/vall show`	Summary table: VD ID, RAID Type, State (Optl, Dgrd, PdFail), Total Size, Cache Policy.
Deep VD Diagnostics	`storcli /c0/v0 show all`	Stripe size, Bad Blocks Table presence, Read/Write Policy (WB/WT), Access policy (RW/RO).
Start Consistency Check (CC)	`storcli /c0/v0 start cc`	Scans parity data across array and fixes silent corruption. Crucial monthly maintenance.
Monitor CC Progress	`storcli /c0/v0 show cc`	Shows % completed and estimated time remaining for consistency check.
Start VD Initialization	`storcli /c0/v0 start init [full]`	Initializes new virtual drive. Fast init by default; full init writes zeros to entire array.
Delete Virtual Drive	`storcli /c0/v0 del [force]`	Destroys virtual drive volume (ALL DATA PERMANENTLY ERASED).

Target RAID Configuration	Creation Command Syntax (Direct, WriteBack, ReadAhead)
RAID 0 (Stripe)	`storcli /c0 add vd type=raid0 drives=252:0,252:1 [Direct] [WB] [ra]`
RAID 1 (Mirror)	`storcli /c0 add vd type=raid1 drives=252:0,252:1`
RAID 5 (Single Parity)	`storcli /c0 add vd type=raid5 drives=252:0-4 pdperarray=5`
RAID 6 (Dual Parity)	`storcli /c0 add vd type=raid6 drives=252:0-5 pdperarray=6`
RAID 10 (Striped Mirror)	`storcli /c0 add vd type=raid10 drives=252:0-3 pdperarray=2`

6. Hot Spares & Rebuild Management

Operation	Command	Details & Engineering Notes
Assign Global Hot Spare	`storcli /c0/e252/s4 add hotsparedrive`	Designates standby spare for any degraded RAID array on controller 0.
Assign Dedicated Hot Spare	`storcli /c0/e252/s4 add hotsparedrive dgv=0`	Binds spare drive exclusively to Drive Group 0.
Start Manual Rebuild	`storcli /c0/e252/s1 start rebuild`	Initiates RAID parity reconstruction onto replacement drive in Slot 1.
Monitor Rebuild Progress	`storcli /c0/e252/s1 show rebuild`	Displays rebuild % completion, elapsed time, and ETA.
Set Rebuild Priority Rate	`storcli /c0 set rebuildrate=40`	Sets rebuild bandwidth percentage (1-100%). Default is 30%.

7. Foreign Configuration Diagnostics & Array Migration

Step / Action	Command Syntax	Impact & Recovery Procedure
1. Detect Foreign Arrays	`storcli /c0/fall show`	Scans all slots and reports any drives holding foreign RAID metadata.
2. Preview Foreign Layout	`storcli /c0/fall preview`	Simulates what virtual drives will be reconstructed before committing import.
3. Import Foreign Array	`storcli /c0/fall import`	**DATA RECOVERY:** Imports foreign metadata and restores access to the Virtual Drives.
4. Clear Foreign Metadata	`storcli /c0/fall clear`	**WIPES RAID HEADER:** Erases foreign metadata to reuse the disks as fresh spares.

8. Advanced Cache Policy Tuning & Performance Optimization

Tuning controller cache algorithms maximizes IOPS and throughput while ensuring data safety.

Cache Setting	Command Syntax	Performance & Safety Trade-off
Write-Back Cache	`storcli /c0/v0 set wrcache=WB`	Writes acknowledge immediately in controller RAM. Maximum performance (Requires BBU/CV).
Write-Through Cache	`storcli /c0/v0 set wrcache=WT`	Writes acknowledge only after reaching physical disk. Safe without BBU, slower write speed.
Read Ahead (RA)	`storcli /c0/v0 set rdcache=RA`	Reads sequential blocks into controller cache. Accelerates sequential streaming read workloads.
Direct / Cached I/O	`storcli /c0/v0 set iopolicy=Direct`	**Direct:** Bypasses cache buffer for random small blocks. **Cached:** Buffers all reads in RAM.
Disk On-Board Cache	`storcli /c0/v0 set pdcache=on|off`	Enables/Disables drive internal volatile RAM cache. Must be OFF unless protected by UPS.

9. Online Expansion, RAID Migration & Secure Disk Sanitization

Operation	Command Syntax	Operational Function
Online Capacity Expansion	`storcli /c0/v0 expand size=500`	Expands virtual drive size live by 500GB without taking array offline.
RAID Level Migration (RLM)	`storcli /c0/v0 start migrate type=raid5 option=add drives=252:2`	Converts RAID 1 to RAID 5 live by adding disk 252:2 on the fly.
Monitor Migration Progress	`storcli /c0/v0 show migrate`	Displays live % progress for RAID Level Migration or OCE expansion.
Instant Crypto Erase (SED)	`storcli /c0/eX/sX start erase crypto`	Cryptographically wipes Self-Encrypting Drive (SED) instantaneously.
Full Zero Wipe / Sanitize	`storcli /c0/eX/sX start erase thorough`	Overwrites disk sectors with patterns for secure decommissioning (DoD compliant).
Monitor Erase Progress	`storcli /c0/eX/sX show erase`	Displays disk zero-wipe % progress and estimated completion time.

10. Enclosure Environmental Monitoring & Config Backups

Feature / Task	Command Syntax	Diagnostic Purpose
Enclosure Sensor Temps	`storcli /c0/eall show temp`	Displays backplane temperature sensors and overheating alerts.
Backplane Fan RPMs	`storcli /c0/eall show fan`	Monitors cooling fan tachometers and detects failed fan modules.
Power Supply Status (PSU)	`storcli /c0/eall show psu`	Displays redundant power supply AC input health and failure alerts.
Backup Controller Config	`storcli /c0 dump file=ctrl0_backup.xml`	Exports entire RAID configuration and drive mappings to XML backup.
Restore Controller Config	`storcli /c0 restore file=ctrl0_backup.xml`	Restores array settings onto a replacement controller card.
Set System Boot Drive	`storcli /c0/v0 set bootdrive=on`	Configures Virtual Drive 0 as primary BIOS boot volume.

11. Patrol Read & CacheVault / Battery Backup (BBU)

Component	Command	Purpose & Maintenance Practice
Patrol Read Status	`storcli /c0 show patrolread`	Displays Patrol Read status (Active/Stopped), mode, and schedule.
Start Patrol Read	`storcli /c0 start patrolread`	Forces an immediate background surface sector scan of all member physical disks.
CacheVault Status (CV)	`storcli /c0/cv show all`	Displays flash/supercapacitor state: Temperature, Voltage, Learn cycle, SOH.
BBU Status (Battery)	`storcli /c0/bbu show all`	Displays battery charge level, replacement flags, and calibration health.
Start BBU Learn Cycle	`storcli /c0/bbu start learn`	Calibrates battery capacity. (Note: Controller drops to Write-Through cache during learn).

12. Incident Response & Troubleshooting Runbooks

Runbook A: Replacing a Failed Drive in a Degraded Array

1. Identify failed disk slot: storcli /c0/eall/sall show (Look for Failed or UBad)
2. Blink chassis LED: storcli /c0/e252/s2 start locate | **3. Pull bad disk and insert replacement**
4. Stop locator LED: storcli /c0/e252/s2 stop locate | **5. If UBad:** storcli /c0/e252/s2 set good force
6. If Foreign: storcli /c0/fall clear | **7. Start rebuild:** storcli /c0/e252/s2 start rebuild | **8. Monitor:** storcli /c0/e252/s2 show rebuild

Runbook B: Proactive Replacement on Predictive Failure (PdFail)

1. Locate drive: storcli /c0/eall/sall show (State: PdFail) | **2. Take offline:** storcli /c0/e252/s1 set offline
3. Mark missing: storcli /c0/e252/s1 set missing | **4. Spin down:** storcli /c0/e252/s1 spin down
5. Physical replacement | **6. Set Good:** storcli /c0/e252/s1 set good force | **7. Start Rebuild:** storcli /c0/e252/s1 start rebuild

13. Automated Health Audit Script (storcli_admin_diag.sh)

An automated administration & diagnostics tool is included in this repository: storcli_admin_diag.sh.
- **Run interactive management menu:** sudo ./storcli_admin_diag.sh
- **Automated cron health audit:** sudo ./storcli_admin_diag.sh --audit (Generates timestamped log in ./storcli_reports/)
- **Chassis drive locate:** sudo ./storcli_admin_diag.sh --locate 0 252 1 | **Silence alarm:** sudo ./storcli_admin_diag.sh --silence 0

14. State Acronyms & Diagnostic Status Reference

Code	Full Name	Severity	Meaning & Required Engineering Action
Onln	Online	**NORMAL**	Physical drive is functioning normally within a Virtual Drive.
Optl	Optimal	**NORMAL**	Virtual drive has all member disks healthy with full redundancy.
UGood	Unconfigured Good	**STANDBY**	Healthy drive not part of any array; ready for allocation/spare.
UBad	Unconfigured Bad	**ERROR**	Drive has errors or unrecognized config. Run set good force.
Dgrd	Degraded	**WARNING**	Virtual drive has 1+ failed drives. Operating on parity. Rebuild ASAP.
PdFail	Predictive Failure	**WARNING**	SMART error threshold exceeded. Replace disk before fatal crash.
Rbld	Rebuilding	**ACTIVE**	Parity data is actively being reconstructed onto replacement disk.
Offln	Offline	**ERROR**	Drive or array taken offline. Virtual drive is inaccessible if offline.
Frgn	Foreign	**NOTICE**	Drive holds RAID metadata from another controller. Import or Clear.
JBOD	Just a Bunch of Disks	**RAW MODE**	Raw pass-through mode bypassing RAID engine (used for ZFS/Ceph).